University of New Haven

POMPEA COLLEGE OF BUSINESS

Marketing

Course Information

MKTG 6640-02
Marketing Analytics
Fall 2023
Credit Hours: 3

Faculty Contact Information

Instructor: Dr. Hesam Teymouri Semester: Fall 2023
Email: XXX
Classroom: X
Class Meeting Time: Tuesdays 2:30 pm - 5:15 pm

COURSE SYLLABUS

Course Description

Prerequisites: **BANL 6100, MKTG 6610**. This course prepares students for careers in marketing analytics by developing the ability to analyze data using statistical and computational tools to support informed decision making. Topics include methods used in online marketing, grocery stores, and financial markets. The course also explores customer big-data techniques and their theoretical foundations to help students acquire practical skills through hands-on experiences. Students are provided with the tools and the confidence necessary to analyze real-world questions and present their findings. **3 credits.**

Course Introduction:

Businesses depend on customers to create value and sustain growth. Customer analytics refers to data-driven approaches that use customer information to guide decisions related to acquiring, serving, developing, and retaining customers. In practice, however, customer data rarely provide a complete picture of the market. Important elements—such as customer preferences, willingness to pay, or the likelihood that a customer may discontinue using a product—are often not directly observable in available datasets. In addition, datasets typically exclude individuals who could become potential customers but have not yet interacted with the firm. As a result, effective customer analytics combines data, marketing theory, domain knowledge, and statistical models to generate meaningful insights and support business decision making.

The primary objective of this course is to help students understand how analytics can be used to support data-driven marketing decisions. Students will learn how to interpret analytical methods, apply statistical techniques, and work with code used in data analysis. Throughout the course,

students will use R, an open-source programming language widely used for statistical analysis, data visualization, and modeling. The course emphasizes hands-on learning, based on the idea that analytical concepts are best understood when they are implemented and explored through coding and practical data analysis. Because marketing analytics is commonly conducted in collaborative teams in industry, students will also work in groups on several analytical projects where they jointly analyze datasets and interpret results.

Students will work with R, an open-source programming language widely used for statistical analysis, data visualization, and modeling. The course will use base R along with packages from the Tidyverse, a widely adopted collection of tools designed for data manipulation, visualization, and analytical workflows. These tools are commonly used in both academic research and industry because they support efficient and collaborative data analysis.

Students often ask whether they should focus on learning R or Python for analytics. Both languages are widely used and share overlapping capabilities, although each has particular strengths. Python is a general-purpose programming language that is widely used for data engineering, large-scale data processing, and machine learning. R, in contrast, was developed specifically for statistical analysis and data visualization, making it particularly well suited for modeling and exploratory data analysis. Because this course focuses on visualization and analytical modeling, R is a natural tool for many of the tasks we will perform. In practice, many analysts benefit from familiarity with both languages, and developing the ability to learn new analytical tools is an important professional skill in data-driven careers.

Enrollment in the course is an agreement to abide by and accept all terms. Remember that you may elect to drop or withdraw from this course before the end of the drop/add period.

Recommended Text(s)/Materials/Supplies

[Course GitHub address should be provided here]
PowerPoint Slides, Assigned readings (available on Canvas)

Course Goals

Develop the ability to use marketing analytics to answer key business questions. The course uses analytical tools such as Python, R, and Excel to analyze marketing data. This course takes a practical and quantitative approach to marketing analytics from the perspective of a business analytics professional. The students will learn the most relevant techniques used in the real world by data analysts of companies around the world.

Course Learning Objectives

- Apply data reduction techniques to appropriate data for a real-world question
- Analyze relevant data using appropriate analytical methods in R
- Identify appropriate data sources to address real-world business questions
- Apply the techniques of social listening and content analysis
- Present and communicate analytical findings from marketing data both individually and as part of a team.

Course Requirements & Assessment

A. INSTRUCTIONAL METHODOLOGY:

The course will consist of lectures, in-class exercises, and hands-on data analysis using R and Excel. Students will work with real datasets and complete a series of applied analytics projects throughout the semester. Several of these projects will be completed in teams to reflect how marketing analytics is conducted in professional settings, where analysts collaborate to interpret data and generate insights. Examinations and a final project will assess students' understanding of the analytical methods and their ability to apply them to real-world problems.

B. ATTENDANCE:

This is an on-ground class and consistent attendance is expected. The course outline shows that you will miss the grades of in-class activities if you skip the sessions. Also, for every unexcused absence, 20 pts will be taken off your final grade. Following is the attendance rule:

If you have a medical condition that results in more than two absences and want the absences to be excused, you must provide documentation to the Dean of Students Office, who will verify your condition and notify me.

Please do not request to attend on Zoom. Unless you have a permission from the Accessibility Resource Center online attendance is not permitted.

C. COURSE REQUIREMENTS:

Assignments: Students will complete a total of 8 group assignments throughout the term that allow them to practice the analytical concepts introduced in the course. These assignments involve applying analytical techniques to real datasets using tools such as R and Excel. While the assignments are completed in teams, each student is expected to contribute meaningfully to the analytical work and demonstrate individual understanding of the methods used. Overall, assignments carry 30% of your final grade. Assignment materials will be posted on Canvas.

Make-up Assignments: No make-up assignments will be given.

Online Certificates: TBA

Examinations: Midterms and Final Exam: We will use multiple-choice and short-answer questions to test your understanding and ability to apply key ideas. All exams will be on paper without access to digital resources, to disincentivize over-reliance on language models. You may bring any printed or written material you want. Readings, class material, scripts and homework questions will all be covered. We will not ask you to produce original code during the exams, but we will expect you to understand and interpret relevant code in the scripts, and we will ask questions that require reasonable understanding of the homework exercises. This will all be within reason; we will not intentionally write trick questions. The midterms will be during class in weeks 4 and 8. The final exam will be in person at the university-appointed time and place. If a student misses any exam without a documented, university-sanctioned reason or exam alternative then we will set their exam score to 5 points below the minimum exam score in the section. Please plan ahead to ensure you do not miss the exams. Students need to use their computers to answer the questions.

Make-up Examinations: If you cannot take an exam, you must inform me about your absence before the exam. Moreover, you must have a VALID reason for not taking the exam on the scheduled date such as family emergencies and school affiliated activities. You need

to provide me with the valid documentation as well regarding your absence, and I will schedule a new date for you to take the exam. The make-up exam must be taken no later than 1 week after the main scheduled exam.

Expectations

- Ability to self-discipline (time, etc.) and to learn through various instructional methodology
- You need to bring your own laptop if there is no computer in the classroom
- Basic mathematical/statistical knowledge
- Ability to work cooperatively in team-based analytical projects and group class activities
- Ability to use the statistical software R and R Studio
- Ability to use Microsoft Excel and have an active version of Microsoft Excel installed on your laptops

This course will require significant in-class and out-of-class commitment from each student. The University estimates that a student should expect to spend two hours outside of class for each hour they are in a class. (For example, a three-credit course would average six [6] hours of additional work outside of class).

Grading

Grades earned are based on your performance on homework, quizzes, exams and the final exam.

Gradable Component	Description	Points
Final Exam (Individual)	Comprehensive exam evaluating students' understanding of analytical methods and their ability to interpret marketing data and model results.	250
Midterm Exam (Individual)	Exam assessing students' understanding of foundational analytics concepts including data preparation, exploratory analysis, and segmentation methods.	200
Individual Analytics Exercise	Individual assignment where students analyze a dataset and interpret analytical results independently. This ensures each student demonstrates their own analytical skills.	150
Team Analytics Projects (8 total)	Team-based projects applying analytical techniques such as exploratory data analysis, clustering, RFM analysis, predictive modeling, and market basket analysis to real marketing datasets.	300
Team Contribution (Peer Evaluation)	Team members evaluate each other's participation, analytical contribution, and collaboration within group projects. This component ensures accountability in team-based analytical work.	100
Total		1000

Late submissions will get 20% reduction penalty. Do not wait for the last day to complete and submit your assignments.

Final Grades are assigned with the following scale:

Undergraduate Scale

Grades Scored Between	Letter Equivalents
93 to 100	A
90 to Less than 93	A-
87 to Less than 90	B+
83 to Less than 87	B
80 to Less than 83	B-
77 to Less than 80	C+
73 to Less than 77	C
70 to Less than 73	C-
67 to Less than 70	D+
63 to Less than 67	D
60 to Less than 63	D-

Graduate Scale

Grades Scored Between	Letter Equivalents
93 to 100	A
90 to Less than 93	A-
87 to Less than 90	B+
83 to Less than 87	B
80 to Less than 83	B-
77 to Less than 80	C+
73 to Less than 77	C
70 to Less than 73	C-
Less than 70	F

Course Outline/Schedule

Week 1

Introduction to Marketing Analytics & Data in Marketing

- Role of analytics in marketing decision making
- Types of marketing data (transactional, behavioral, survey, digital)
- Structured vs unstructured data
- Overview of analytics workflow
- Intro to R environment

Week 2

Data Preparation & Data Quality

- Data types (categorical, numerical, time series)
- Data cleaning

- Missing data handling
- Feature creation
- Data transformations and scaling

Week 3

Descriptive Statistics & Exploratory Data Analysis

- Summary statistics
- Distribution analysis
- Correlation analysis
- Data visualization for marketing data
- Identifying patterns in customer data

Week 4

Exploratory Analysis for Customer Behavior & STP

- Behavioral data exploration
- Customer heterogeneity
- Segmentation, Targeting, Positioning framework
- Preparing data for segmentation

Week 5

Customer Segmentation & Clustering Analysis I

- Distance metrics
- Hierarchical clustering
- Dendrogram interpretation
- Segment interpretation

Week 6

Customer Segmentation & Clustering Analysis II

- Principal Component Analysis (PCA)
- Dimensionality reduction
- K-means clustering
- Segment validation

Week 7: Midterm

Week 8: Fall Break

Week 9

Recency, Frequency, Monetary (RFM) Analysis

- Behavioral segmentation
- Customer value analysis
- Managerial interpretation

Week 10

Predictive Modeling & Customer Churn

- Introduction to predictive analytics
- Logistic regression intuition
- Churn prediction
- Model interpretation

Week 11

Choice Modeling & Marketing Experiments

- Choice-based conjoint analysis
- A/B testing
- Experimental design
- Statistical testing

Week 12

Marketing Attribution & Conversion Analytics

• Conversion funnels
• Attribution models
• Multi-channel marketing analytics
• Campaign performance measurement

Week 13

Customer Lifetime Value (CLV)

- CLV calculation methods
- Retention modeling
- Marketing strategy implications

Week 14

Association Rules & Market Basket Analysis

- Support, confidence, lift
- Apriori algorithm
- Retail analytics applications

Week 15: Final Exam

Due to unforeseen circumstances, it may be necessary to alter the outlined schedule of activities, assignments, tests, quizzes and presentations etc. during the semester. Any such changes will be clearly communicated to the class.

Diversity Statement

The University of New Haven embraces diversity and recognizes our responsibility to foster a diverse, inclusive, and welcoming environment in which all members of the Charger community of all backgrounds and identities can learn, work, and live together. We benefit from the academic, social, and cultural developments that arise from a diverse campus that is committed to equity, inclusion, belonging, and accountability.

We have a responsibility as a community and as individuals to address and remove barriers, achieve success, and sustain a culture of inclusivity, empathy, kindness, and compassion. We encourage, welcome, and embrace participation in ongoing dialogue, engagement, and education to critically examine and thoughtfully respond to the changing realities of our community. Diversity, equity, inclusion, acceptance, and belonging enrich the Charger community and are instrumental to institutional success and fulfillment of the University mission.

Reporting Bias Incidents

At the University of New Haven, there is an expectation that all community members are committed to creating and supporting a climate which promotes civility, mutual respect, and open-mindedness. There also exists an understanding that with the freedom of expression comes the responsibility to support community members' right to live and work in an environment free from harassment and fear. It is expected that all members of the University community will engage in anti-bias behavior and refrain from actions that intimidate, humiliate, or demean persons or groups or that undermine their security or self-esteem.

If you have an immediate safety concern for yourself or others, and/or believe someone poses an immediate threat to themselves or others, please contact University Police at 203-932-7070 or call 911. Community members can report bias-motivated incidents by completing the form at **www.newhaven.edu/biasreporting**. Community members are encouraged to complete this form if they are the target of bias or harassing behaviors, witness such behaviors, or gain knowledge of these behaviors occurring within the University community. All matters concerning bias and harassment will be handled by the Dean of Students Office and Human Resources Office.

University-wide Academic Policies

A continually-updated list of University-wide academic policies and descriptions of key university student resources, can be found on Canvas. You can access them by simply clicking on the (?) help button.

The University-wide academic policies include (but are not limited to) the University's attendance policy, procedures for both adding / dropping a course and course withdrawals, an explanation for the sorts of circumstances where incomplete (INC) grades could be considered by the faculty, and the academic integrity policy (among others). Also in this location you will find information regarding the process for reporting bias and topics related to our maintaining a

positive learning environment (including, but not limited to, discrimination and sexual misconduct).

The list of key university student resources to enable learning include (but are not limited to) the University's Center for Student Success, Writing Center, Center for Learning Resources, and the Accessibility Resource Center.

Course Delivery Options (choose the statement that describes the modality of *your* course).
For courses with a location of ONLI that also list time and day information, students should plan to be available during that time for synchronous online learning. Classes with no time listed are asynchronous, fully online, and coded ONLI. Due to social distancing requirements, some courses will be offered in Hybrid or Flex formats. ***Online learning may occur at any time, depending on how the University may determine classes are best taught under any circumstance.*** *All courses, including selected labs, practicums, project-based, and clinical courses may require limited, additional meeting times to accommodate accreditation, regulatory, and similar requirements and students will be advised about this at the beginning of those classes.*

Hybrid: Students attend in-person for half of the scheduled course time and the other half of the class content is available online. Students are only required to attend in-person on their designated day(s) and students **need not attend classes virtually**.

Online asynchronous (ONLI): Fully online asynchronous course with **no live, required class sessions**.

Remote synchronous (ONLI with specific meeting time): Fully online synchronous course. **Students attend virtually at the assigned class time.**

Flex: Half of the students attend each class session in-person and the other half attends virtually at the same time.

On-Ground: Fully on-ground course with every student meeting in-person.

Clinical/Laboratory: Practical courses that have an on-ground required component.

UNIVERSITY STUDENT SUPPORT SERVICES

The University recognizes that students can often use some help outside of class and offers academic assistance through several offices.

Accessibility Resources Center

Students with disabilities, chronic health-related conditions, or military service-connected disorders are encouraged to share, in confidence, information about course specific approved reasonable accommodations. The Accessibility Resources Center, located in Sheffield Hall, is responsible for and committed to providing supports and

resources that serve to promote educational equity and ensure that students are able to participate in the opportunities available at the University of New Haven. Reasonable accommodations are not made without written documentation from the Accessibility Resources Center.
The Accessibility Resource Center can be reached at (203) 932-7332 or by email at AccessibilityResCtr@newhaven.edu. For additional information, please refer to the Accessibility Resource Center (ARC) website at www.newhaven.edu/campusaccess. For additional assistance from the Dean of Students Office, please contact: deanofstudents@newhaven.edu. If you require assistance with the technology requirement, please visit the Student Technical Support page.

Center for Learning Resources (CLR)
The Center for Learning Resources (CLR), located in the Peterson Library, provides academic content support to the students of the University of New Haven using metacognitive strategies that help students become aware of and learn to apply optimal learning processes in the pursuit of creating independent learners. CLR tutors focus sessions on discussions of concepts and processes and typically use external examples to help students grasp and apply the material.

Center for Student Success (CSS)
The Center for Student Success provides students with a multitude of resources available on campus and assists students in fulfilling their educational, social and personal goals.

Counseling & Psychological Services (CAPS)
CAPS offers confidential, free services in order to support student mental health and wellbeing. The services include individual and group therapy, support groups, consultations, and 24/7 crisis support. We are available in person and remotely, and are in the office M-F, 8:30-4:30. Please call us to schedule an appointment or with any questions at 203-932-7333; you can also schedule online. If you experience a mental health crisis after hours, you can call our main number for support.

Myatt Center
The Myatt Center for Diversity and Inclusion is committed to creating a multicultural environment through intentional education, campus community engagement, and valuing the unique identities of each member of the Charger Community. Our commitment to diversity is driven by the core values of connection, belonging, inclusivity, equity, acceptance, and accountability. The Myatt Center's focus is to create a respectful and inclusive environment based our awareness and ability to engage with others who are different on many levels including ethnicity, race, sexual orientation, gender, military, religious belief, and life experiences.

Marvin K. Peterson Library
The Library provides access to online databases, e-books, e-journals, electronic U.S. Government Documents, print books, educational games, and audiovisual materials. A search can be conducted through all these resources at once by using the search box "Articles, Books, & More."

The Library provides three floors with individual quiet study space, collaborative group study space, study rooms with technology, whiteboards, Dell desktops, iMacs, scanners, and printers. The entire library is a wireless zone.

Librarians assist in locating relevant sources of information for research papers, thesis, hon ors thesis, and other projects. Librarians answer general reference questions and help with effectively evaluating sources of information. Help is available through a Chat Service , 24/7 Ask a Librarian Service, a Zoom Reference Service, and by E-Mail. Complete the Research Consultation Form to arrange a time convenient for you.

LibGuides are created
to assist students with research. They contain an overview of resources available through the libr ary, as well as tutorials, subject guides, and course specific guides.

University Writing Center

The mission of the Writing Center is to provide high-quality tutoring to undergraduate and graduate students as they write for a wide range of purposes and audiences. Tutors are undergraduate and graduate students who are majoring in a variety of fields across the University. We are here to work with you at any stage in the writing process; just bring in your assignment, your ideas, and any writing you've done so far. To make an appointment, you can register for an account with our scheduling site https://newhaven.mywconline.com or visit us in person at our desk on the first floor of Peterson Library (just to the left after you enter the library).